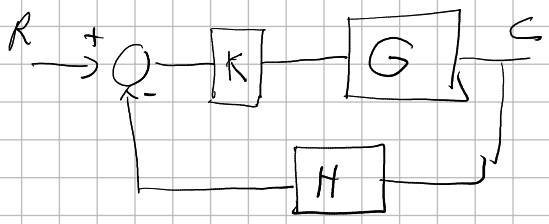
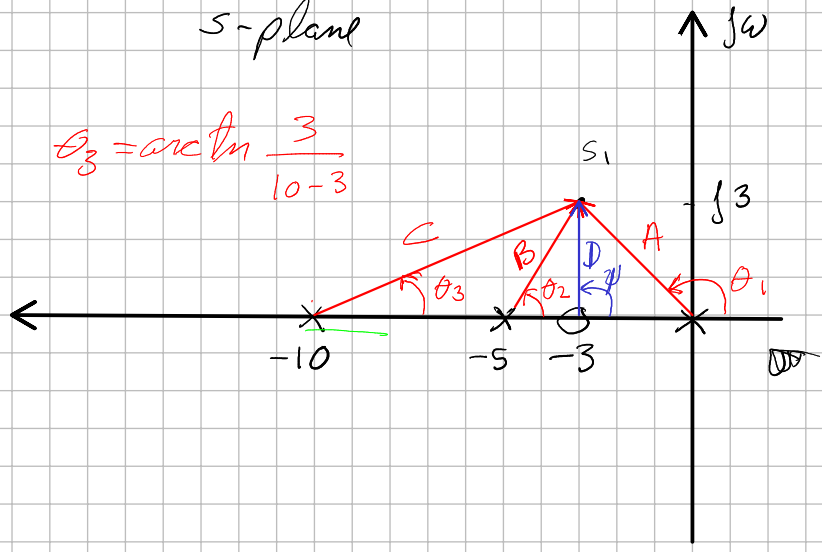
$$GH = \frac{K(s+3)}{s(s+5)(s+10)}$$

open loop

$s_1 = -3+3j$ — does it belong to the root locus?





$$\frac{C}{R} = \frac{KG}{1+KGH}$$

$K: 0 \to \infty$

$1 + KGH = 0$

(poles of the closed loop)

s_1 is a pole of the closed loop system (belongs to the root locus) $\quad l = 0, 1, 2, 3 \ldots$

1) $\angle KGH(s_1) = \angle GH(s_1) \overset{?}{=} \pm 180(2l+1)$

if true, s_1 is a candidate to belong to the rocus (to be a closed loop pole)

2) $|KGH(s_1)| = 1 \Rightarrow$ definitely is a pole of the closed loop system.

$$K = \frac{1}{|GH(s_1)|}$$

1) angle condition $\quad \psi - \theta_1 - \theta_2 - \theta_3 \overset{?}{=} \pm 180(2l+1)$

if (1) is true, then I can find K to satisfy condition (2) (magnitude condition)

$$\left|\frac{K\,D}{ABC}\right| = 1 \qquad K = \frac{ABC}{D}$$

$1 + G(s_1)H(s_1) = 0$ if s_1 is a closed loop pole

$G(s_1)H(s_1) = -1 + j0$

$s_2 = -6.27 + j\,8.23$ ✓ is a closed loop pole if $K = 87.91$